



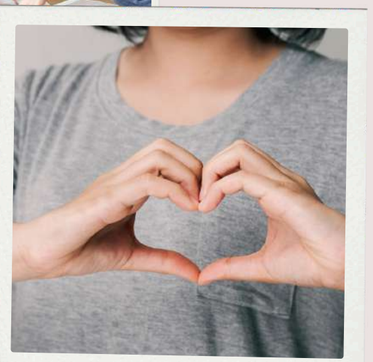





FAMILY PHOTO WALL

family vault

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Introduction



Families today live far apart due to work, studies, and lifestyle changes.
The emotional connection slowly decreases when memories are not shared regularly.
Family Vault provides a secure, private digital space where families can upload, view, and relive memories together.

Problem Statement

- Families experience emotional distance due to geographical separation.
- Traditional photo albums are outdated and difficult to maintain.
- Social media lacks privacy and emotional warmth.
- No dedicated digital space designed specifically for families.

Objectives

- Create a private and secure family-only photo sharing platform.
- Provide a simple and nostalgic photo-album experience.
- Enable easy uploading, viewing, and organizing of memories.
- Strengthen emotional bonding among family members.

Motivation

- Need for a calm, distraction-free space unlike public social media.
- Families want to preserve memories safely.
- Emotional connection increases when memories are shared.
- Inspired by traditional photo albums, redesigned for digital era.

Scope

Current Scope

- Login/Register
- Upload photos
- View shared dashboard
- Create/Join family
- Add captions

Future Scope

- AI photo categorization
- Face Recognition
- Voice messages
- Mobile App
- Shared family event calendar

Use Case Diagram

📸 Upload / View / Delete Photos
Members can share their important moments, browse family memories, or remove unwanted images.

🗂️ Manage Albums
Photos can be grouped into events (like birthdays, trips, festivals) to keep the wall organized and easy to navigate.

✉️ Invite New Members
Admins or existing members can send invitations so relatives living far away can join the wall and stay connected.

🎉 Create Celebration Walls
For special days—like birthdays, anniversaries, and achievements—members can create dedicated celebration spaces.

📝 Add Captions & Tags
Users can describe the memory, add names, or tag family members to make each photo more meaningful and expressive.

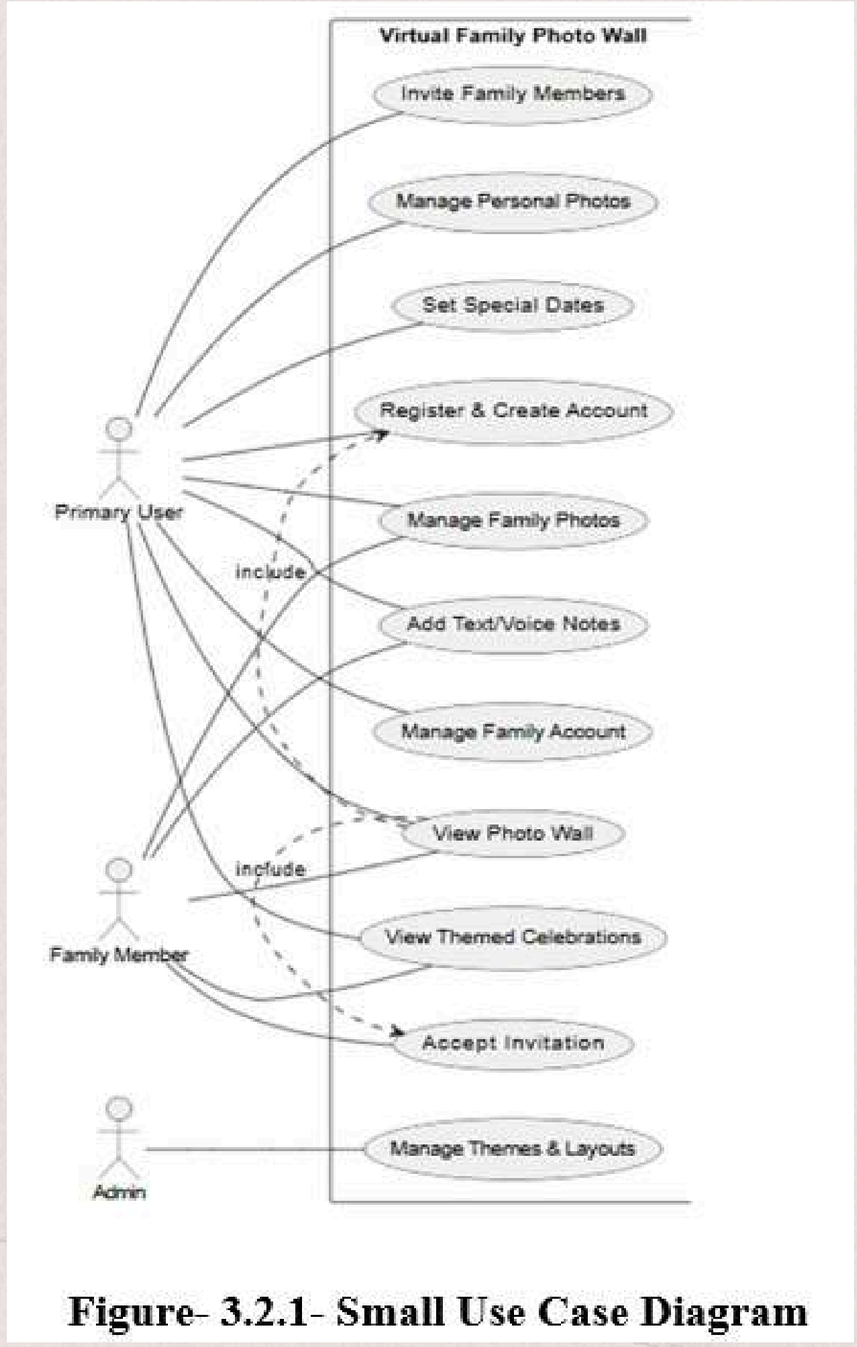
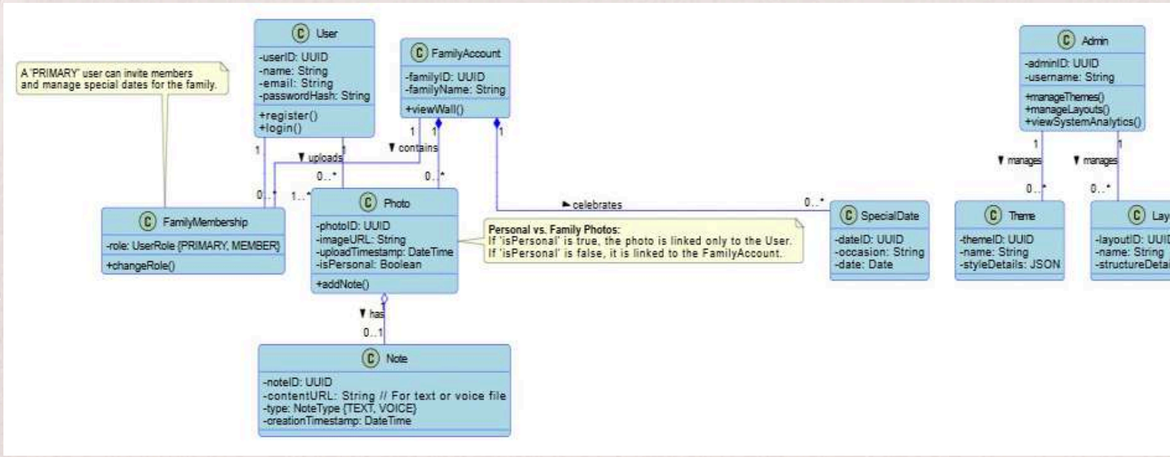


Figure- 3.2.1- Small Use Case Diagram

Class Diagram



- User Class

Stores user details like name, email, and relationship.
Handles login and access to the family wall.

- Photo Class

Contains information about each uploaded photo — caption, date, and who uploaded it.

Used to display memories on the main dashboard.

- Album Class

Groups photos into event-based albums such as birthdays, trips, or festivals.

Helps keep family memories organized.

- Memory Class

Represents a complete memory post with the photo, caption, and interactions. Adds emotional depth to shared moments.

- Database Handler Class

Handles communication with the database. Stores, updates, and retrieves user, photo, and album data.

Activity Diagram

- User logs in

The system verifies the user's credentials and grants access.

- Dashboard loads

The user sees the shared family wall and available options.

- User selects "Upload Photo"

Chooses an image from their device.

- System validates the photo

Checks format, size, and file type.

- Photo is stored in the database

Metadata like caption, user, and date are saved.

- Updated family wall is shown

The new photo appears instantly for all family members.

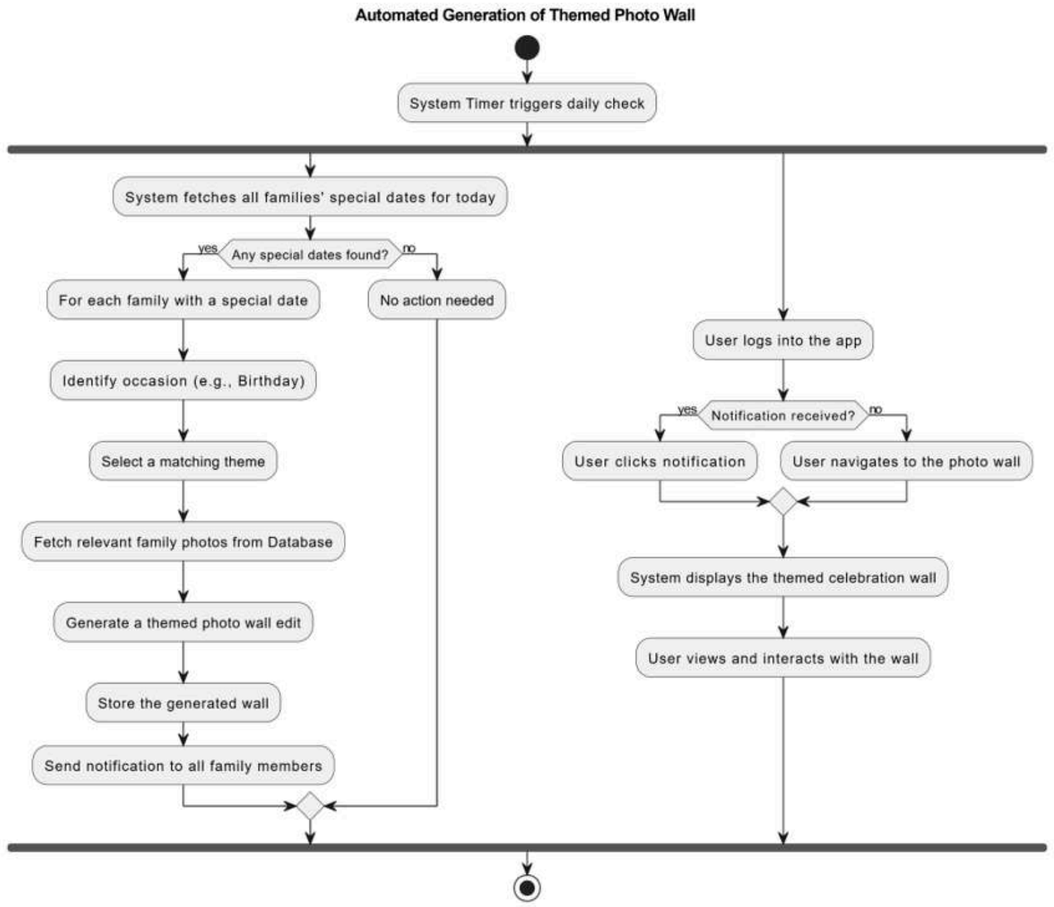


Figure- 3.4.2- Full Activity Diagram

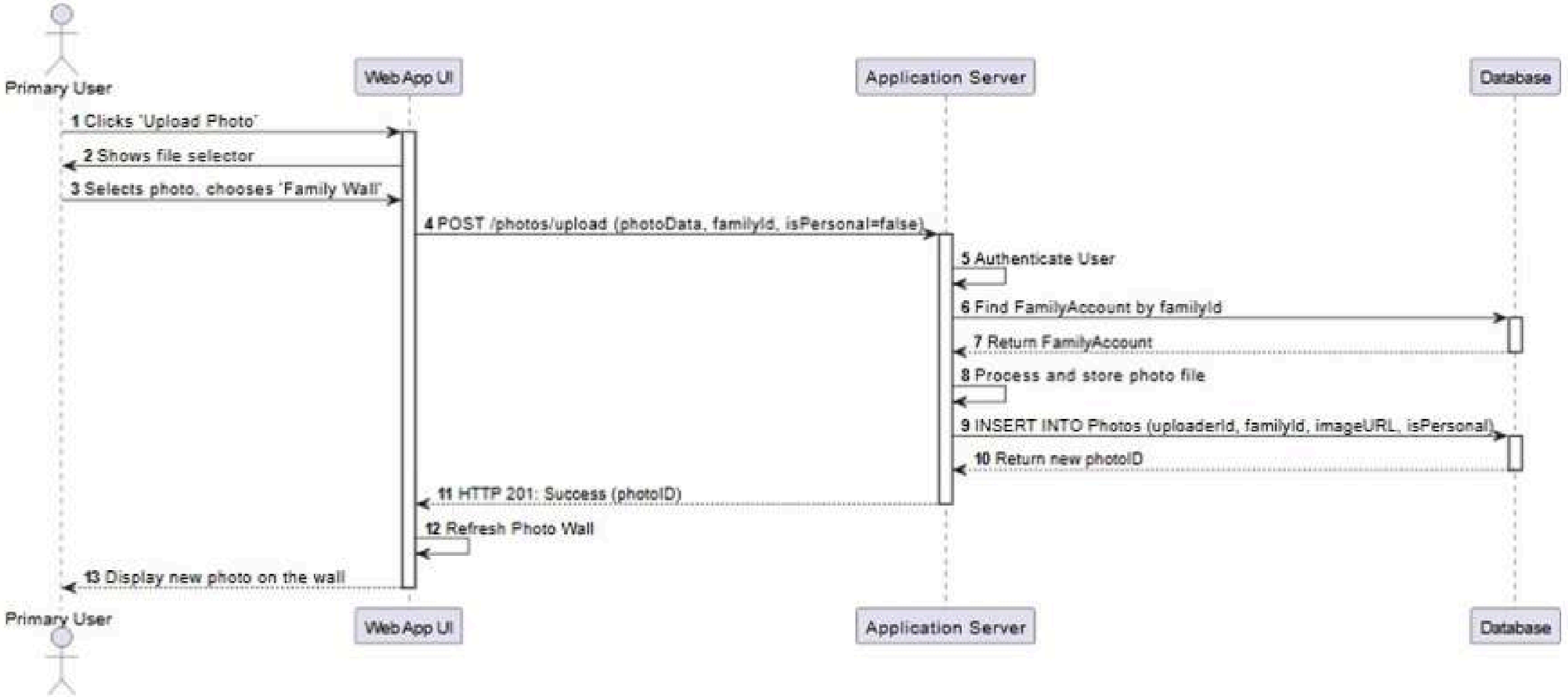
Sequence Diagram

Photo Upload Sequence

- User sends upload request
  Triggers the photo upload action from the interface.
- Interface forwards data to backend
  Photo file and caption are processed.
- Backend validates and prepares the data
  Ensures the photo meets requirements.
- Database stores photo and details
  Image URL, caption, and timestamp are saved.
- Backend sends success message
  User receives confirmation of successful upload.

Invite Member Sequence

- User selects “Invite Member”
  Enters the relative’s email or family ID.
- System sends invitation
  Email or in-app invite is delivered.
- Member accepts the invite
  Joins the family group using the shared ID.
- Database updates member list
  New member is added to the family wall.
- Confirmation shown to user
  Group now reflects the updated member count.


Primary User
Web App UI
Application Server
Database
1 Clicks "Upload Photo"
2 Shows file selector
3 Selects photo, chooses "Family Wall"
4 POST /photos/upload (photoData, familyId, isPersonal=false)
5 Authenticate User
6 Find FamilyAccount by familyId
7 Return FamilyAccount
8 Process and store photo file
9 INSERT INTO Photos (uploaderId, familyId, imageURL, isPersonal)
10 Return new photoID
11 HTTP 201: Success (photoID)
12 Refresh Photo Wall
13 Display new photo on the wall
Primary User
Web App UI
Application Server
Database

Data Flow Diagram

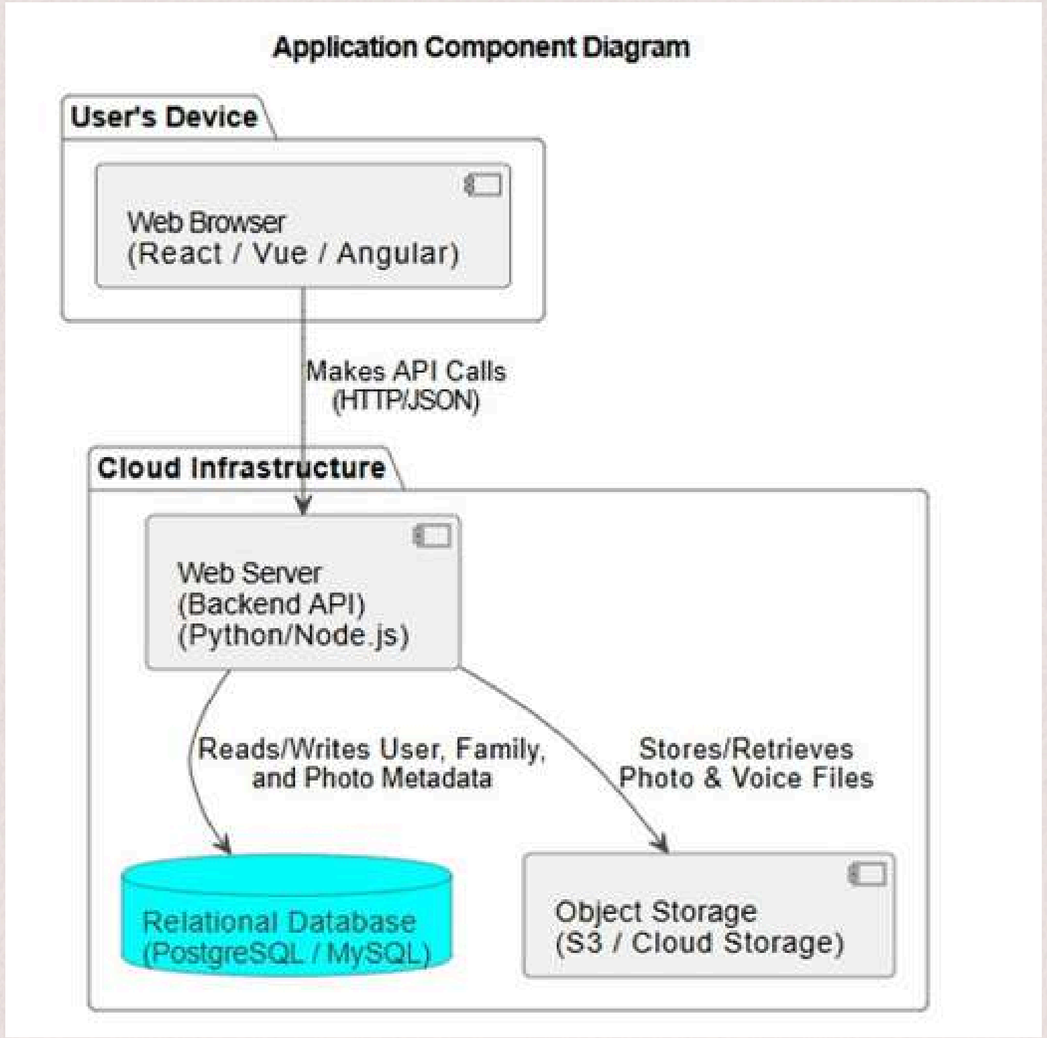
- User Registration & Login
  User enters credentials → system authenticates → access granted.

- Photo Upload Process
  User uploads image → system checks it → database stores photo+caption.

- Album Management
  System organizes photos into albums based on event or date.

- Viewing Memory Wall
  User requests dashboard → system fetches photos → displays family wall.

- Database Operations
  Handles all reading/writing of user data, photos, captions, and albums



System Architecture

- Three-Tier Architecture

The system follows a structured three-layer design to keep the platform organized, scalable, and easy to maintain.

- Frontend (User Interface)

The part of the system that users interact with.
Displays the family wall, photo uploads, albums, and all visuals in a simple and friendly way.

- Backend (Logic & Authentication)

Handles the main operations behind the scenes.
Manages login, processes uploads, communicates with the database, and ensures everything works smoothly.

- Database (Storage Layer)

Stores all important data including photos, captions, user accounts, and albums.
Ensures memories are saved safely and can be retrieved quickly.

- Why This Architecture?

It keeps the system flexible, easy to update, and capable of handling growing users and photos while maintaining smooth performance.

Implementation

- Prototype Designed Using Figma & Canva
  The initial layout, color scheme, and user flow were created using design tools to ensure the platform feels warm and family-friendly.

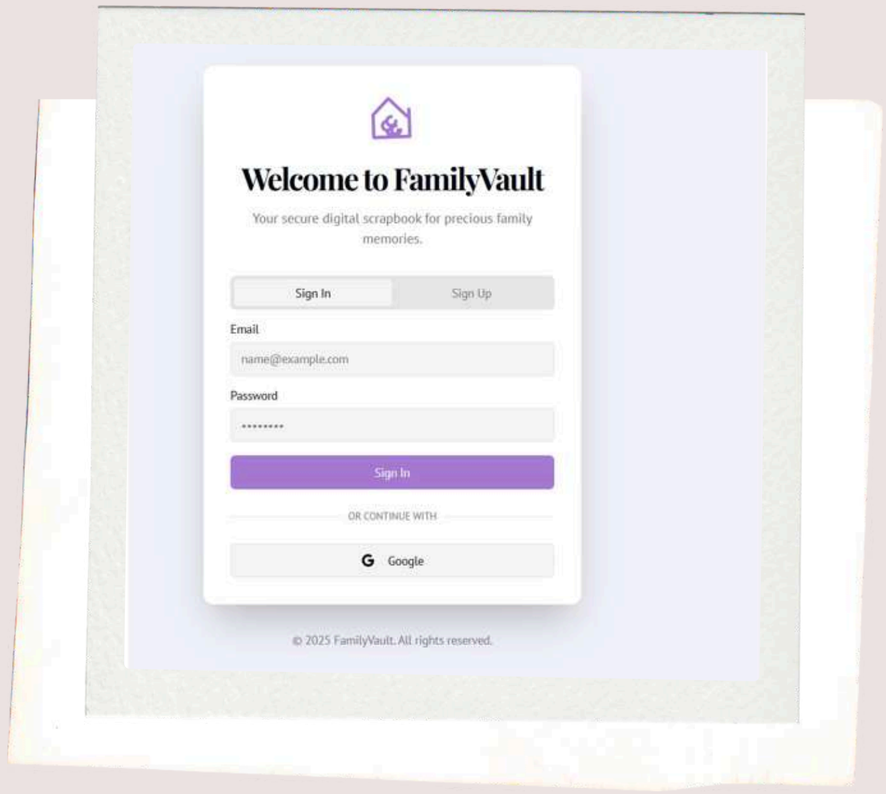
- Core Screens Developed
  Includes Login, Register, Dashboard, and Photo Upload pages — the essential features needed to demonstrate the concept.

- Clean, Warm, and Nostalgic UI
  Soft colors, simple layouts, and a familiar photo-album feel make the interface comfortable for all users.

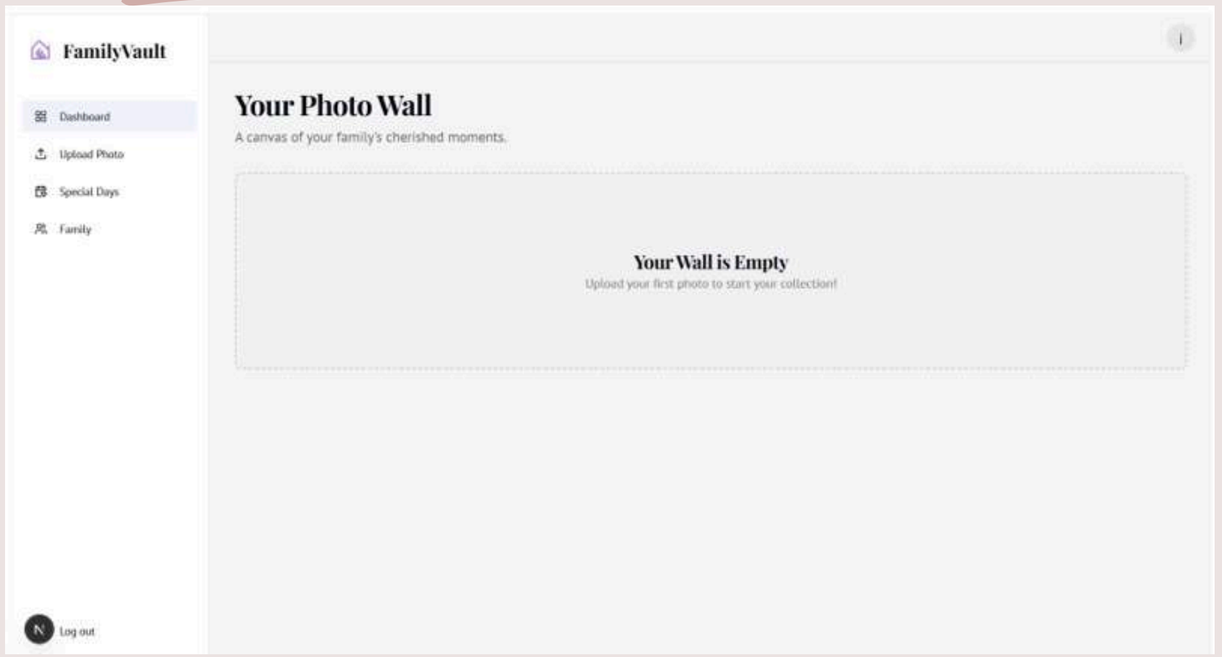
- Easy for All Age Groups
  The design avoids complexity so even children or elderly family members can use it without confusion.

- User Testing & Feedback
  The prototype was shared with users, and feedback was positive — they found the system simple, intuitive, and emotionally engaging.

Results

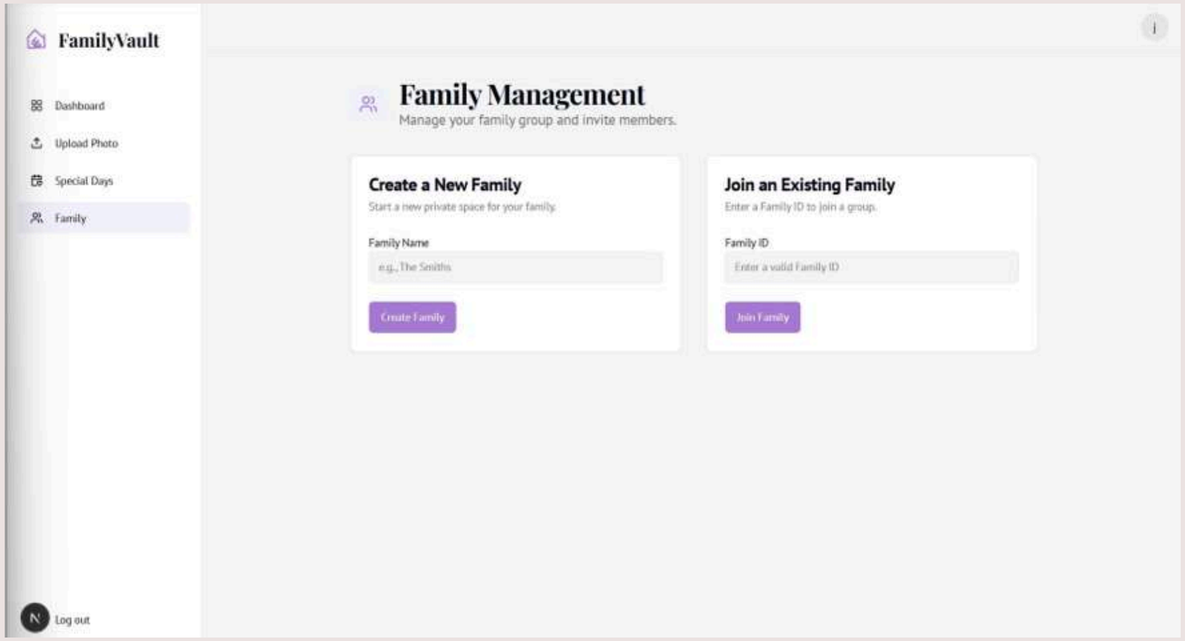


It shows the signin or signup page.

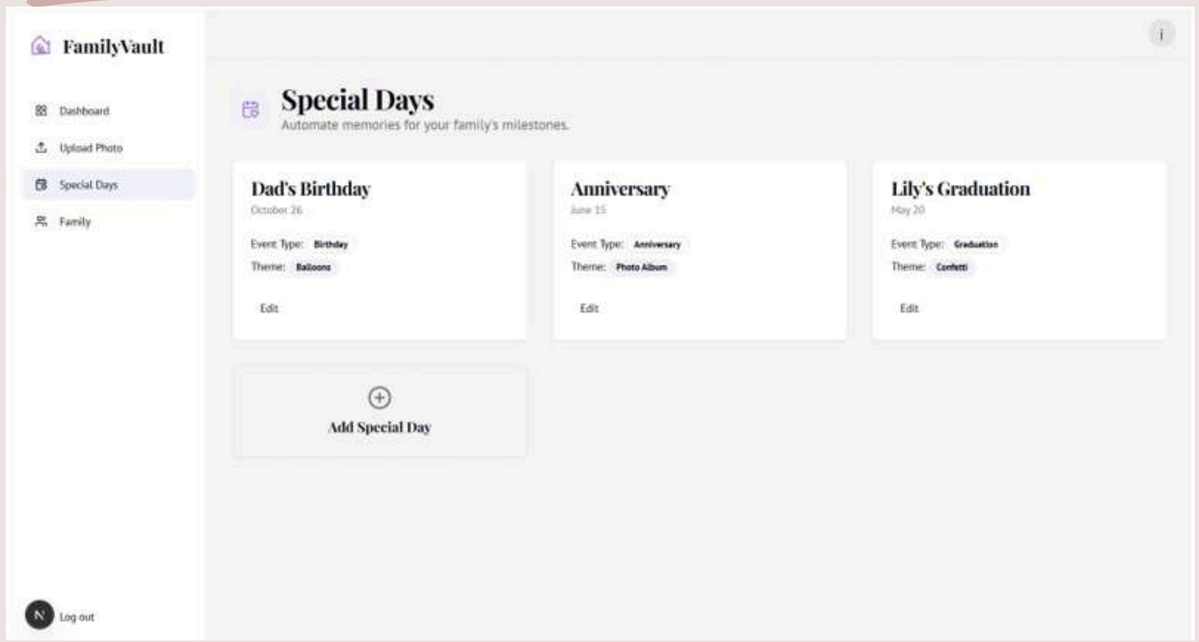


It shows a empty dashborad a person get to this page after logging in

Results



It shows that the person after logging has two options of chossing to create a family or joining a family.



It shows that the family members can add the special days like birthdays,anniversary in the web app.

Conclusion

- Strengthens Family Connections

Family Vault helps bridge emotional distance by allowing members to share and relive memories digitally.

- Simple, Private, and User-Friendly Platform

The design focuses on privacy and ease of use, making it suitable for families of all ages.

- Achieves Core Objectives

Meets the goals of usability, security, emotional engagement, and smooth interaction.

- Potential for Future Growth

Future updates can include AI features, mobile applications, voice notes, event reminders, and more interactive tools.

- A Step Toward Digital Family Bonding

The project shows how technology can preserve emotions and memories while bringing families closer.

Thank You